

UTS ENGSOCC

**Industry Partner Project
Building R13 Development**

AGENDA

- 1 Introduction
- 2 Building R13 Brief
- 3 Site Location
- 4 Core Engineering Problem
- 5 Next Steps

Introduction

James Nelson – Principal Civil Engineer

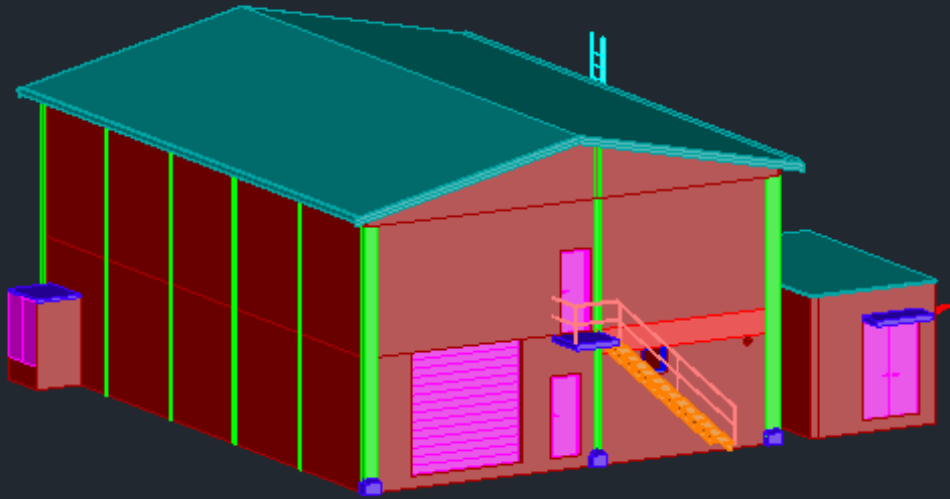
- 9 Years in the Construction Industry
- Varied Project experience
- Initial projects being Sydney Metro USDTs
- Secondment to Victorian Rail Infrastructure Project, completing concept utility design for new rail lines
- Completed Detail Design of SouthWest Sydney Metro
- Lead NSW Ports Division completing 3 concurrent construction projects
- Currently completing a combination of
 - Public Domain works
 - Department of Defence Works Nationally



TASK OVERVIEW



Building R13 Brief



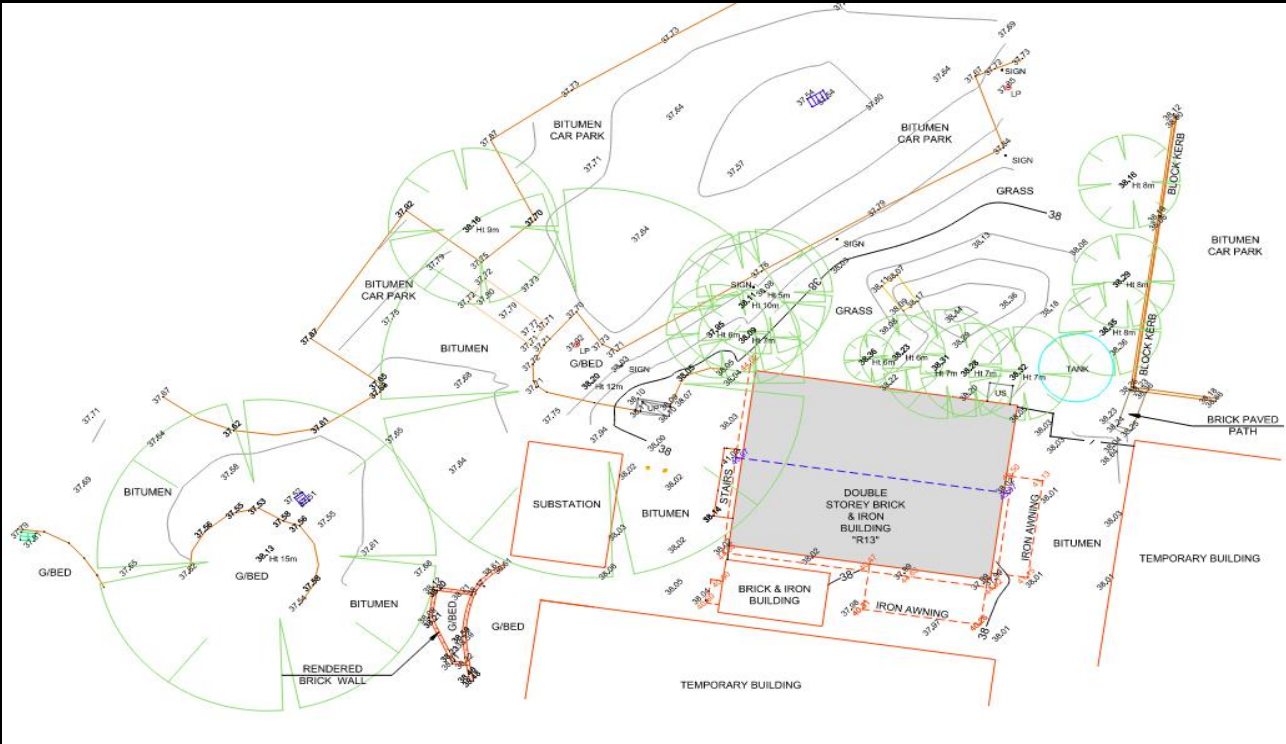
- The R13 Building was previously used as a small car garage and is being converted into a functional base for the Oceanography Team.
- A key component of this conversion was an upgrade to enable outdoor boat washdown, supporting research operations
- This Washdown area is to meet safety, environmental, and regulatory requirements.

Building R13 – Functional Use

- The building is now being converted into a base for the Oceanography Team, which supports fieldwork involving small research vessels. This change of use is central to understanding the engineering problem.
- A car garage typically has minimal water supply requirements, basic drainage, and limited environmental controls
- The campus setting introduces further constraints, including proximity to pedestrian routes, nearby buildings, and existing underground services.
- As engineers, you must recognize that adapting existing assets is often more complex than designing new ones, because the solution must work within inherited constraints. Understanding the history and setting of the R13 Building will help you appreciate why careful consideration of layout, drainage, and environmental controls is required for something as seemingly simple as washing down boats.

Site Location

Survey & Google Map



Existing Site Conditions and Constraints

In its pre-development condition, the R13 site comprised hardstand areas with basic drainage infrastructure and limited-service capacity, consistent with its former use.

The site is spatially constrained with limited expansion opportunity and is surrounded with active campus usage.

The active pedestrian movement, adjacent building structures, and existing underground utilities limits the location for proposed infrastructure.

Existing site topography governs surface runoff direction; therefore, it is essential to understand how runoff is currently captured before proposing any drainage systems.

From an engineering perspective, it is essential to identify existing systems that can support the development where possible, while minimising site reconfiguration that may increase costs, disruption, and approval complexity.

Core Engineering Problem



How can an outdoor boat washdown facility be safely, effectively, and environmentally responsibly integrated into an existing campus site while meeting regulatory requirements?



Water Supply Requirements



Ensuring adequate water pressure, flow, and accessibility is critical for the washdown facility's functionality.



Runoff Water Management



Managing runoff that may contain salt, sediments and contaminant prior to discharging to the existing drainage system.



Safety and Risk Measures



Design must consider any imposing risks to the users and public in the busy campus environment.

YOUR ROLE AS STUDENT CONSULTING ENGINEERS

Adopt an Engineering Mindset

- Approach the project as graduate engineers by carefully interpreting the brief, identifying constraints, and proposing a feasible design solution.

Sound Reasoning and Judgement

- Apply professional judgement by making reasonable assumptions and developing defensible concept-level solutions.
- Detailed hydraulic calculations or drawings are not required.

Effective Communications

Clearly state any assumptions made where information is missing and explain them logically in your presentation.

Consider Alternative Options

- Evaluate possible design options and clearly justify your preferred approach and why it is feasible for the site.



Purpose of the Task

1.

Understand how a change in building use drives new infrastructure requirements

2.

Interpret project constraints (site, stakeholders, environment, regulations)

3.

Describe and justify a civil works solution at a concept level

4.

Communicate engineering ideas clearly to a non-specialist audience

NEXT STEPS AND METHODOLOGY

Review Site Information

Begin by examining all site data, photos, and constraints while discussing insights with your team to establish a common understanding.

Define Problem and Requirements

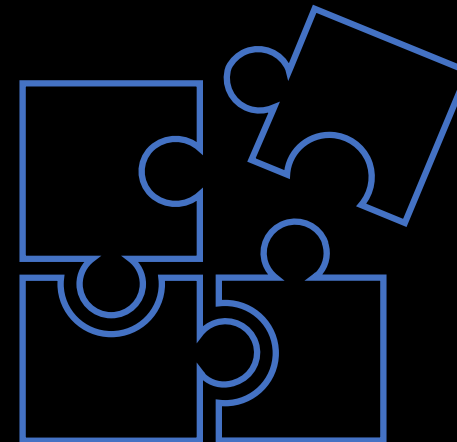
Describe the problem in your own words and set out the key requirements for a solution.

Brainstorm and Manage Uncertainty

Generate layout and drainage ideas early while managing uncertainties through assumptions and identifying any missing information.

Align With Operational Needs

Continuously verify that your solution meets the oceanography team's operational needs and respects campus constraints.



Task Delivery

- **5-Minute Presentation Overview**
- Site Context and Key Constraints
- Key Findings
- Design Assumptions
- Proposed Solution



GOOD LUCK